When Low WER Becomes Dangerous: Counterfactual Decision-Stability ASR for High-Stakes Speech-Driven Decision Systems

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Abstract

Speech-driven decision systems increasingly transform conversations into operational signals for routing, escalation, compliance monitoring, and case-handling decisions. In high-stakes anti-fraud calls, small ASR differences can land on decision atoms such as negation, amount, actor, action, time, and intent. A transcript can therefore remain close under WER or CER while a plausible alternative transcript changes the downstream decision.

We propose Counterfactual Decision-Stability ASR (CDS-ASR), a framework that evaluates whether downstream decisions remain stable under acoustically and semantically plausible transcript alternatives. CDS-ASR extracts risk atoms, constructs plausible ASR alternatives, scores decision instability with Counterfactual Escalation Instability Score (CEIS), and applies constrained recovery or conservative machine action for high-risk spans while preserving aggregate-only reproducibility for sensitive call evidence.

Across a six-model 258-row split comparison, selected-300 high-stakes provenance outputs, and a completed 300-row dual-reviewer human audit with 900 model-level assessments per reviewer, CDS-ASR provides a consequence-centered evidence layer beyond transcript similarity. The dual-reviewer audit records strong agreement for decision-change, semantic-risk, expected-safe-action, and confidence labels, with Cohen's kappa values of 0.849970, 1.000000, 0.851426, and 0.934274, respectively. In the current aggregate predictor and policy-replay surface, CEIS achieves the highest decision-change AUC and a zero-false-negative operating point, while risk-triggered policies, including CEIS-triggered conservative action, eliminate high-risk missed and critical miss counts under the aggregate-only claim boundary. These results support decision-stability evaluation as a scoped companion to transcript accuracy, semantic metrics, confidence-aware correction, and reviewer-visible human-audit reliability.

Introduction

Speech is no longer only a record of what happened in a call. In modern contact centers, speech is increasingly converted into operational signals: categories, summaries, compliance cues, real-time alerts, routing decisions, and escalation support. Commercial contact-center analytics already describe voice conversations and transcripts as inputs for automated categorization, summarization, compliance monitoring, and real-time assistance (Amazon Web Services 2026a, 2026b). This turns ASR output into a decision substrate. A transcript can influence how a case is categorized, which queue receives it, whether an alert is raised, and whether a human reviewer ever sees the full call context.

Anti-fraud calls make this speech-to-decision pipeline concrete. Taiwan's National Police Agency describes the 165 hotline as a public anti-fraud channel where staff record incident details and provide information to victims (National Police Agency, Ministry of the Interior 2024). The operational scale is large: the FBI's 2025 Internet Crime Report context reported more than one million IC3 complaints and large cyber-enabled fraud losses (Federal Bureau of Investigation 2026b, 2026a). In this setting, the important spoken facts are often small: whether money was already transferred, whether the amount was 30,000 or 300,000, who initiated contact, when the event happened, whether the caller is certain, and whether the scenario matches a known scam pattern. These are not merely words. They are decision atoms.

This paper starts from the following practical risk: a transcript can be linguistically close and still operationally unsafe. A single ASR ambiguity around negation, amount, actor, action, time, intent, or scam pattern can move a case from routine review to priority review, or from escalation to unsafe downrouting. In this paper, "dangerous" is operationalized as ASR-induced decision change that can produce unsafe downrouting, high-risk missed escalation, or critical miss under the declared anti-fraud triage policy. The title is therefore a measurable claim about decision consequences, not a general safety slogan.

Existing ASR evaluation and repair methods provide the foundation for this problem. WER and CER remain necessary because they make transcript accuracy auditable and comparable. Semantic ASR metrics make the evidence more task-aware: Kim et al. motivate Semantic Distance by showing that WER can fail to reflect downstream spoken-language-understanding behavior, and Rugayan et al. evaluate Aligned Semantic Distance as a semantic metric that better aligns with human judgments and downstream tasks (Kim et al. 2021; Rugayan, Salvi, and Svendsen 2023). Transcript repair also advances the pipeline. Naderi et al. study post-hoc ASR correction with LLMs and use confidence-based filtering to reduce harmful changes to likely accurate transcripts (Naderi et al. 2024). Selective prediction, reject-option, and conformal prediction work further show how systems can trade coverage, abstention, and uncertainty when acting on uncertain predictions is costly (Chow 1970; Geifman and El-Yaniv 2017, 2019; Angelopoulos and Bates 2021).

These methods make speech systems more measurable, semantic, correctable, and uncertainty-aware. The remaining gap is the decision-stability question. A semantic metric can report that two transcripts are close or far. A correction model can make a transcript more fluent or more likely. A confidence filter can avoid some risky changes. Yet a high-stakes workflow also needs to know whether plausible ASR alternatives would change the downstream action. In anti-fraud triage, the actionable question is not only "How accurate is the transcript?" but also: "Would another plausible transcript lead to a different safe action?"

We propose Counterfactual Decision-Stability ASR (CDS-ASR) to answer that question directly. CDS-ASR extracts decision-critical risk atoms, generates acoustically and semantically plausible transcript alternatives, measures decision instability with Counterfactual Escalation Instability Score (CEIS), and evaluates constrained recovery or conservative machine action for high-risk spans. WER, CER, semantic metrics, SRES, confidence, and correction baselines remain comparison layers. The central contribution is the new evaluation view: ASR for high-stakes speech-driven systems should be judged by whether the downstream decision remains stable under plausible transcript alternatives.

Verified source anchors for this introduction are maintained in `citation_seed.md` and `references.bib`, with access dates refreshed on 2026-05-28. Empirical manuscript claims point to the aggregate artifacts listed in the Appendix.

Related Work

Transcript-Centered ASR Metrics

Transcript accuracy metrics provide the reproducible baseline for ASR comparison. WER and CER remain necessary because they allow stable comparison across models and splits when tokenization, normalization, micro/macro scope, and zero-reference handling are declared. This manuscript keeps WER/CER as auditable surface metrics and uses the repo's metric audit to define the Chinese ASR reporting policy. CDS-ASR builds on that surface layer by asking which transcript differences reach decision atoms and change safe action.

Semantic And Downstream-Aware ASR Evaluation

Semantic ASR metrics extend transcript evaluation toward meaning preservation and downstream task behavior. Kim et al. introduce Semantic Distance for ASR performance analysis toward spoken-language understanding, motivating the move from literal correctness to semantic correctness for intent recognition, slot filling, semantic parsing, and named entity recognition (Kim et al. 2021). Rugayan et al. evaluate Aligned Semantic Distance against WER and show its value for human perception and downstream NLP tasks (Rugayan, Salvi, and Svendsen 2023). CDS-ASR uses this line of work as a foundation and adds a direct decision-stability target: whether plausible ASR alternatives change a high-stakes downstream decision.

Transcript Repair And Conservative Decision Action

LLM and confidence-aware correction methods provide transcript-repair baselines. Naderi et al. show that confidence-based filtering can guide post-hoc LLM correction of ASR transcripts and reduce the chance of changing likely accurate transcripts (Naderi et al. 2024). CDS-ASR complements this line by evaluating the decision interval that remains when plausible ASR alternatives affect risk atoms. Recovery in this manuscript is automatic and machine-bounded: human review supplies evaluation labels and governance evidence, while risk-triggered policies supply the scoped intervention test.

The conservative-action layer also connects to selective prediction and reject-option research. Chow formalizes the error-reject tradeoff for recognition systems (Chow 1970), while modern selective classification and SelectiveNet show how neural systems can abstain or reject to manage risk-coverage tradeoffs (Geifman and El-Yaniv 2017, 2019). Conformal prediction further motivates distribution-free uncertainty sets for high-risk systems (Angelopoulos and Bates 2021). CDS-ASR applies this governance intuition to speech-driven decision evidence: when transcript alternatives affect decision atoms, the system should expose the instability or take conservative action rather than treat one transcript as fully stable.

High-stakes speech domains reinforce the need for consequence-aware ASR evaluation. In psychotherapy, Miner et al. show that ASR feasibility depends on the clinical use case and that individual-level safety monitoring requires a more cautious threshold than population-level language analysis (Miner et al. 2020). CDS-ASR brings the same claim-evidence discipline to anti-fraud calls by separating transcript accuracy, semantic risk, decision instability, and aggregate recovery evidence.

This paper positions ASR models as evidence-producing components. The protagonist is the downstream decision that changes when an acoustically plausible transcript difference lands on a decision-critical atom.

Method

Problem Formulation

We study speech-driven decision systems where an audio segment is transcribed by ASR and then used by a downstream escalation workflow. The downstream label space includes routine review and higher-priority escalation states. The central unit is a model-sample: one ASR hypothesis for one audio row, evaluated against aggregate risk and decision-stability evidence.

The paper asks whether the decision remains stable under plausible transcript alternatives. This makes transcript accuracy a supporting layer and decision-stability the primary high-stakes safety target.

CDS-ASR Pipeline

CDS-ASR evaluates high-stakes ASR through a decision-stability pipeline:

```
audio
-> ASR transcript and runtime/quality signals
-> risk atom extraction
-> plausible counterfactual transcript variants
-> downstream decision model
-> SRES and CEIS scoring
-> constrained recovery or conservative machine action
```

Risk atoms are transcript spans whose alternatives can affect downstream decisions. This manuscript focuses on negation, amount, action, actor, time, intent, uncertainty, and scam-pattern atoms.

CEIS scores the maximum decision-flip risk among plausible variants. In this paper-facing implementation, the plausibility term is a bounded proxy rather than a calibrated acoustic posterior:

```
CEIS(x) = max over v in V(x) [
    Plausibility(v | x) * RiskAtomWeight(v) * DecisionDistance(f(x), f(v))
]
```

`Plausibility(v | x)` denotes a bounded proxy plausibility score derived from model disagreement, Mandarin phonetic ambiguity, domain-slot alternatives, and available ASR/runtime signals. `RiskAtomWeight(v)` is a risk-atom class weight defined by the decision-critical atom schema. `DecisionDistance` maps the downstream label space into an ordinal safety distance over `no_escalation`, `review`, `priority_review`, `critical_escalation`, conservative machine action, and abstention, with critical misses treated as the highest-risk direction for conservative action.

Downstream Decision Function

The downstream decision function `f` is fixed before CEIS scoring. It maps each transcript or plausible variant into a declared triage action:

```
f(transcript) in {
    no_escalation,
    manual_review,
    priority_review,
    critical_escalation,
    conservative_machine_action,
    abstain
}
```

The distance function is policy-aligned rather than language-model-generated: same-action pairs have distance 0, neighboring review/escalation states have distance 1, and movement from no escalation or routine handling to critical escalation receives the largest distance. Unsafe downrouting of a critical event receives the maximum penalty used by the policy replay. The method contract is versioned in `docs/downstream_decision_contract.md` and `80_semantic_risk_asr/scoring/ceis_config.json` so CEIS rankings can be reconstructed from aggregate artifacts.

CEIS Scale Discipline

All CEIS components are treated as bounded comparison terms. The `Plausibility(v | x)` term represents proxy plausibility derived from model disagreement, Mandarin phonetic ambiguity, domain-slot alternatives, and available runtime signals. `RiskAtomWeight(v)` and `DecisionDistance(f(x), f(v))` provide the risk-atom and decision-distance handles that make ablation interpretable. The submission hardening plan therefore treats ablation as a direct contribution test: CEIS full, CEIS without atom weights, CEIS without plausibility, and CEIS binary atom. These analyses should be regenerated from aggregate-safe reviewed evidence before final CSL submission, and reported as planned validation if the current artifact package has not yet regenerated the table.

Thresholds reported in the current predictor table are diagnostic operating points selected on the scoped reviewed audit. They are useful for aggregate comparison and reviewer inspection, but they are not frozen deployment thresholds unless later selected on a separate development set.

Recovery remains automatic and machine-bounded. Human review is used as an evaluation and governance layer, not as the proposed recovery method. The recovery policies compare no recovery, confidence-only trigger, SRES-triggered recovery, CEIS-triggered conservative action, and CEIS ensemble arbitration.

Risk Atom Schema

The risk atom schema focuses on decision-critical spans:

Risk atom	Decision role	Example effect
Negation	reverses event state	transfer vs no transfer
Amount	changes severity	30,000 vs 300,000
Action	changes case stage	asking vs reporting vs transferring
Actor	changes scam interpretation	bank, police, family, customer service
Time	changes urgency	today, yesterday, next week
Intent	changes routing	inquiry, report, complaint
Uncertainty	changes confidence and safe action	sure, maybe, not sure
Scam pattern	changes case type	investment, fake police, installment cancellation

Counterfactual Variant Contract

Counterfactual variants are plausible ASR alternatives concentrated around risk atoms. They are not generic paraphrases. A variant can be supported by acoustic ambiguity, Mandarin phonetic confusion, domain-slot alternatives, or model disagreement. The paper-facing claim is not that every possible variant is enumerated; it is that the generated alternatives make downstream decision-instability measurable.

Recovery Policy Contract

The five-policy recovery experiment is:

1. no recovery;
2. confidence-only trigger;
3. SRES-triggered recovery;
4. CEIS-triggered conservative action;
5. CEIS ensemble arbitration.

The scoped recovery claim is evaluated on aggregate human-reviewed selected-300 evidence. Per-row transcript-bearing details stay local. At the policy layer, both SRES-triggered recovery and CEIS-triggered conservative action are reported as risk-triggered conservative policies; CEIS's distinct contribution is evaluated primarily in the predictor layer, the decision-stability framing, and the planned ablation suite over atom weights, plausibility, and binary atom variants.

Reproducibility Layer

The paper-facing ASR surface metric is `cer_zh_micro`. The supplemental word metric is `wer_zh_jieba_micro`. Raw whitespace WER and legacy stored WER fields are audit-only because they are unstable for unsegmented Chinese transcripts.

The metric policy is validated by `70_experiments/runs/wer_metric_audit_2026_05_25/`, which records manifest checks, zero-reference-unit checks, tokenizer and normalization policy, and `jiwer` cross-checks.

The target transcription locale is Taiwan Traditional Chinese. New candidate models must satisfy the strict Taiwan Traditional Chinese locale gate before promotion. Previously completed comparable baselines are retained only as disclosed baselines, with locale-violation counts reported and no promotion claim attached. Post-decode OpenCC or other conversion can be evaluated only as a deployment repair lane; it cannot be used to claim that the raw ASR model passed the locale gate.

Selected-300 human audit evidence is tracked through aggregate outputs. The local transcript-bearing reviewer sheets, merged sheets, and completed package remain outside the committed release boundary. The reviewed scope is risk atoms, decision-change labels, expected safe action, confidence, per-model assessment fields, and per-row timing. The current aggregate status is dual-reviewer complete: reviewer 1 and reviewer 2 each completed 300/300 row-level records and 900/900 model-level assessments. Cohen's kappa is 0.849970 for decision-change, 1.000000 for semantic-risk label, 0.851426 for expected safe action, and 0.934274 for annotation confidence.

All paper-facing claims point to aggregate artifacts. Raw audio, raw transcripts, selected row IDs, hypotheses, reviewer notes, and model weights remain governed local materials.

Evaluation Units And N-Ladder

The manuscript separates evidence units explicitly so that model assessments are not mistaken for independent audio rows.

Layer	Unit	N	Role
Test split	audio rows	258	ASR model comparison
Selected high-stakes provenance	selected candidate rows / outputs	300	row-selection and provenance
Human-reviewed audit rows	audio rows	300	dual-reviewer decision-critical review unit
Human-reviewed model assessments	model-row assessments	900 per reviewer	reviewer agreement and full selected-300 audit evidence

Because model assessments are clustered within selected audio rows, inferential uncertainty should be estimated with row-clustered bootstrap resampling or leave-one-row-out sensitivity analysis. The completed 300-row dual-reviewer audit strengthens the reliability layer; any predictor or recovery table regenerated from the full audit should keep the row-clustered design rather than treating model-level assessments as independent calls.

Selection Provenance And Enrichment

The selected-300 audit surface is an enriched high-stakes sample, not a prevalence-preserving sample of all anti-fraud calls. The selection summary uses aggregate-safe provenance signals from SRES scoring, CEIS scoring, and downstream escalation decisions, including high-risk missed, critical miss, unsafe down-routing, model disagreement, high proxy risk, risk-score fill, and clean-control strata. The aggregate selection thresholds recorded for the current package are `low_wer_threshold=10.0`, `sres_threshold=20.0`, and `ceis_threshold=5.0`.

Predictor results therefore support metric behavior on an enriched high-stakes audit surface rather than population-level risk prevalence. If a submission emphasizes CEIS/SRES separation, a sensitivity check should exclude rows directly selected by CEIS or SRES family signals and confirm whether the same directional pattern remains.

Variant Coverage And Human Review Reliability

The current aggregate-only coverage audit reports CEIS top-atom proxy coverage on the existing replay surface. It records aggregate proxy observations with top-atom counts of negation 47, amount 37, action 3, and actor 3. The completed selected-300 dual-reviewer audit provides the final submission validation layer, while the reviewed selection surface also covers risk-signal atoms in aggregate: action 25, actor 15, amount 23, negation 14, and scam pattern 23 selected rows. These counts support submission-safe coverage auditing, not release of variant text. Source-specific coverage is available for model-disagreement provenance in aggregate; phonetic, domain-slot, runtime-signal, rejected-variant, and full variant-generation logs are not currently reconstructed as release artifacts. Stronger generator claims therefore require a future aggregate variant-generation log.

Human labels now include a completed dual-reviewer audit across the full selected-300 surface. Both reviewers completed all row-level and model-level fields, required fields have no empty values, row-level `yes` decisions have at least one model-level `yes` or `uncertain`, critical atoms are contained in risk atoms, and `yes` rows have expected safe action other than `none`. The agreement layer records Cohen's kappa of 0.849970 for decision-change, 1.000000 for semantic-risk label, 0.851426 for expected safe action, and 0.934274 for confidence. This turns human audit reliability from a planned extension into a reviewer-visible contribution.

Experiments

Experiment 1 evaluates comparable ASR model evidence on the canonical 258-row test split. This table supports model-comparison and split evidence, not the paper-grade selected-300 risk/recovery claims.

Experiment 2 uses selected-300 high-stakes proxy outputs as input provenance. The selected-300 proxy outputs identify an enriched high-stakes evidence surface and the review sample, but their raw rows and transcript-bearing metric inputs remain local or ignored. This experiment supports selection provenance, not population prevalence.

Experiment 3 evaluates WER, CER, SRES, and CEIS against human-reviewed decision-change labels from the selected-300 audit. The latest human audit gate is complete at 300 rows with 900 model-level assessments per reviewer and dual-reviewer agreement summaries. The current predictor table reports the existing aggregate replay surface and should be regenerated from the completed audit before final CSL submission so the full selected-300 reliability layer and CEIS ablations are reflected in the final numbers.

Experiment 4 evaluates five recovery policies against the same human-reviewed evidence layer. This is aggregate policy replay evidence for conservative action and recovery governance; a live deployment trial is a later validation layer.

Candidate ASR and multimodal models are kept in a separate exploratory lane. They must not enter the main ASR table until they pass field-contract, runtime-validity, and Taiwan Traditional Chinese locale gates in order.

Results

The Results section separates comparable ASR evidence, candidate-lane boundaries, human-reviewed predictor evidence, and human-reviewed recovery evidence. This separation preserves claim-evidence alignment: the main ASR benchmark supports model-comparison context, while selected-300 human-reviewed outputs support paper-grade risk and recovery claims.

Human review supplies evaluation labels only. The proposed recovery policies are automatic, aggregate-evaluated policies; no transcript-bearing row content is required for paper-facing claims.

Main Results Tables And Figures

The main article uses a reduced figure and table set. The complete R-generated figure package remains available for reproducibility, but the body emphasizes method, core predictor evidence, risk mechanism, and policy replay.

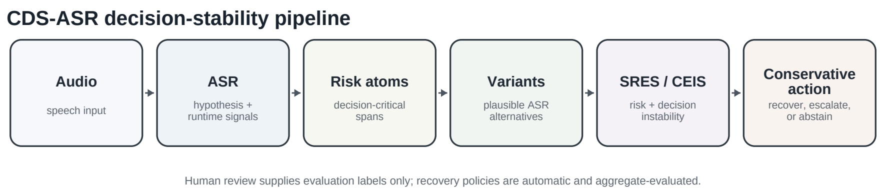

Figure 1: CDS-ASR converts audio into ASR hypotheses and runtime signals, extracts risk atoms, generates plausible variants, scores SRES/CEIS, and applies conservative action. Source: `method section`.

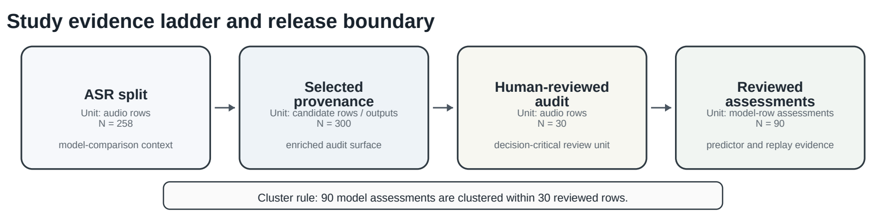

Figure 2: The study separates ASR split evidence, selected-300 provenance, the completed 300-row dual-reviewer audit, and 900 model-level assessments per reviewer. Source: `method evidence units and publishable evidence summary`.

Table 1 reports comparable 258-row split evidence. Run identifiers are kept out of the main table and remain in the aggregate source artifact. The table supports ASR model-comparison context only; final risk and recovery claims are tied to the selected-300 human-reviewed evidence.

Table 3 reports the current aggregate predictor comparison against human-reviewed `human_decision_change_yes` labels. WER/CER are transcript-surface baselines; SRES is the semantic-risk baseline; CEIS is the proposed decision-stability metric. The completed 300-row dual-reviewer audit is the upgraded final-submission evidence surface, so Table 3 should be regenerated with the full audit and the CEIS ablation suite before CSL submission.

CEIS has the strongest point-estimate human-reviewed decision-change AUC and reaches recall 1.0 at the diagnostic threshold, while SRES achieves the highest best-threshold F1 and fewer false positives. This is the paper's central contribution pattern: CEIS exposes conservative decision instability, SRES remains a strong semantic-risk baseline, and the full selected-300 audit plus ablations provide the next reviewer-facing validation layer.

Table 1: Main ASR benchmark on the canonical 258-row split. Run identifiers are reported in the supplement.

Model	Rows	CER zh micro	WER zh jieba micro	Unsafe downrouting	High-risk missed	Locale violations	Paper role
Breeze-ASR-25 partial encoder	258	15.04	21.53	7	4	0	Strongest ASR evidence layer
Breeze-ASR-25 LoRA	258	18.23	25.59	10	7	0	Split/model context
Breeze-ASR-25 base	258	22.72	30.39	34	30	0	Split/model context
Breeze-ASR-26	258	24.27	32.29	27	22	0	Split/model context
Whisper large-v2	258	24.72	32.23	33	28	1	Comparable baseline
Whisper small	258	34.86	43.44	76	70	4	Comparable baseline

CER zh micro is the primary ASR surface metric; WER zh jieba micro is supplemental. Unsafe downrouting and high-risk missed counts are split/model-comparison evidence, not final selected-300 human-reviewed risk claims.

Table 2: Predictor performance against human-reviewed decision-change labels. Unit: 90 model assessments clustered within 30 reviewed audio rows.

Predictor	AUC	AUC CI	Threshold	Best F1	F1 CI	Precision	Recall	FN	Role
WER	0.696	0.570–0.812	42.59	0.462	0.286–0.667	0.391	0.562	7	Surface baseline
CER	0.728	0.608–0.837	16.42	0.452	0.316–0.667	0.304	0.875	2	Surface baseline
SRES	0.899	0.812–0.966	270.00	0.651	0.467–0.842	0.518	0.875	2	Semantic-risk baseline
CEIS	0.912	0.852–0.962	5.00	0.627	0.468–0.815	0.457	1.000	0	Decision-stability metric

Thresholds are diagnostic operating points selected on the scoped audit set, not frozen deployment thresholds. CEIS has the highest point-estimate AUC and zero false negatives at the diagnostic threshold; SRES has the highest best-threshold F1.

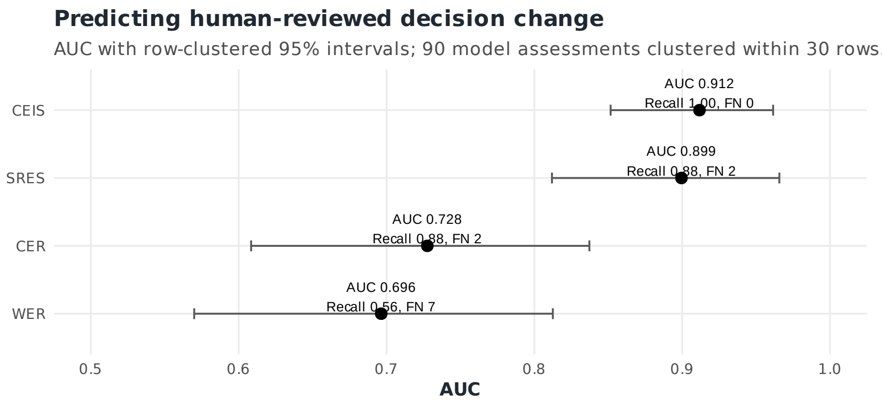

Figure 3: CEIS has the highest point-estimate decision-change AUC and reaches zero false negatives at the diagnostic operating point; row-clustered intervals remain overlapping with SRES. Source: `human_audit_predictor_comparison.tsv` and `human_audit_predictor_clustered_ci.tsv`.

Table 3: Aggregate policy replay under human-reviewed selected-300 labels.

Policy	Unsafe	High-risk	Critical	Budget	Budget CI	Severe elim.	Severe rem. CI	Trig. / elim.
No recovery	29	6	1	0.000	0.000–0.000	0	2–13	n/a
SRES recovery	24	0	0	0.389	0.300–0.489	7	0–0	5.0
CEIS conservative action	24	0	0	0.389	0.300–0.489	7	0–0	5.0
CEIS ensemble arbitration	24	0	0	0.522	0.400–0.656	7	0–0	6.7

Confidence-only recovery is not shown as a main policy because calibrated confidence was unavailable and produced no triggers. Residual unsafe downrouting remains after risk-triggered policies and should be treated as a separate governance issue.

Without recovery, the reviewed evidence contains 6 high-risk misses and 1 critical miss. SRES-triggered recovery and CEIS-triggered conservative action both eliminate high-risk missed and critical miss counts in aggregate policy replay at the same 0.3889 trigger budget, corresponding to 35 triggers for 7 severe missed outcomes eliminated. CEIS ensemble arbitration preserves this 0/0 result while introducing abstention behavior at a higher 0.5222 budget. The policy replay eliminates the most severe missed-risk outcomes under the scoped labels, while residual unsafe downrouting remains at 24 and requires separate governance.

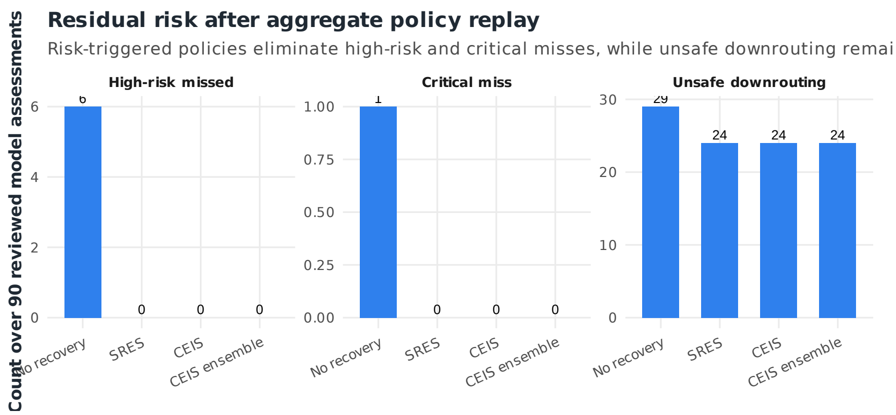

Figure 4: Risk-triggered policies eliminate high-risk missed and critical miss counts in aggregate replay, while residual unsafe downrouting remains. Source: `policy_comparison.tsv`.

A fixed-budget replay provides an additional operating view. At 10%, 20%, 30%, and 40% requested trigger budgets, CEIS-ranked conservative replay leaves 0 high-risk misses and 0 critical misses under the scoped labels; the 10% point uses 9 triggers. SRES-ranked conservative replay leaves 4, 2, 2, and 0 severe missed outcomes at the same requested budgets, reaching 0 severe misses only when all 35 eligible SRES triggers are used. This frontier supports CEIS as a conservative decision-stability signal while preserving the Table 4 result that the diagnostic SRES and CEIS policies tie at the selected 0.3889 budget. Appendix Table A1 reports the fixed-budget replay explicitly as an operating-point frontier rather than a deployment-threshold claim.

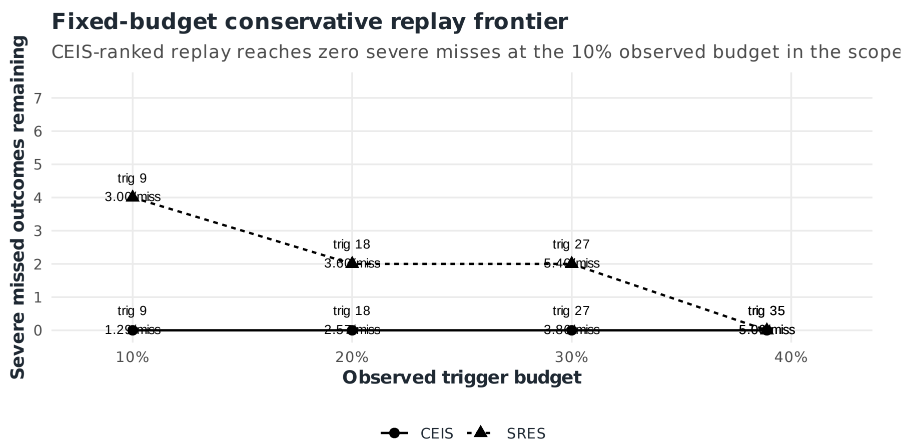

Figure 5: Fixed-budget policy replay shows the trigger-budget tradeoff needed to eliminate severe missed outcomes under scoped labels. Source: `fixed_budget_recovery_frontier.tsv`.

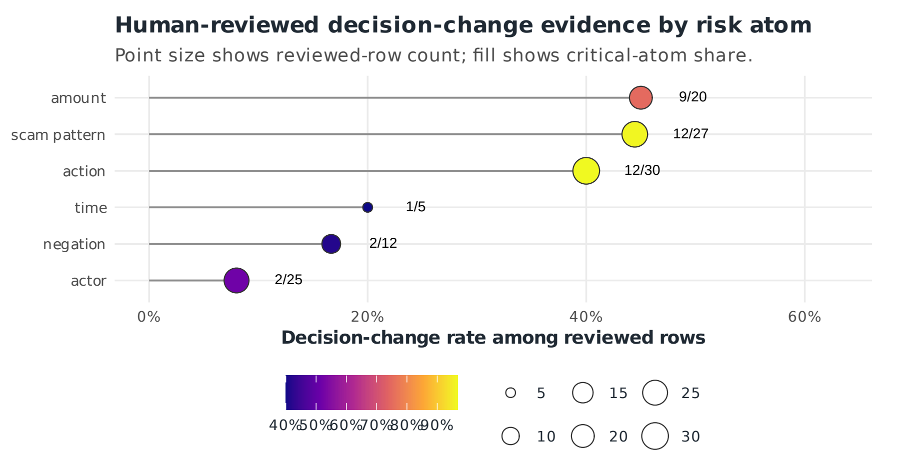

Figure 6: Human review connects risk-atom coverage to criticality and decision-change evidence across reviewed rows. Source: `human_audit_risk_atom_review.tsv`.

Word-level confusion entropy is a valuable next analysis, especially for frequently misrecognized decision words around negation, amount, transfer action, actor identity, and scam-pattern cues. The current submission package keeps that analysis out of the public figure set unless it is first converted into a redacted aggregate lexicon with no transcript spans, audio IDs, selected row IDs, hypotheses, or reviewer notes. A future local-only confusion lexicon can support a supplemental word-level entropy heatmap after privacy review.

Discussion

The evidence supports a consequence-centered ASR evaluation claim: correct WER/CER reporting is necessary, but transcript similarity alone does not capture high-stakes decision instability. The reviewed selected-300 predictor table shows that SRES and CEIS align more strongly with human-reviewed decision-change labels than WER/CER in this scoped evidence layer.

The model comparison should be presented as an evidence layer, not as a model leaderboard. The partial encoder is the strongest current ASR hypothesis generator on the comparable split, and the selected-300 human-reviewed evidence shows how downstream decision evidence refines transcript-centered model comparison.

The recovery result supports a scoped replay claim. Under human-reviewed selected-300 labels, risk-triggered conservative policies, including CEIS-triggered conservative action, eliminate high-risk missed and critical miss counts in aggregate replay. SRES-triggered recovery and CEIS-triggered conservative action reach this result at the same 0.3889 trigger budget, while ensemble arbitration adds interval/abstention behavior as a higher-budget governance option. CEIS is therefore best presented as the more conservative decision-instability signal at the predictor layer, while SRES remains a strong semantic-risk recovery baseline.

Improving ASR remains necessary, but stronger transcript models do not replace decision-stability analysis. The paper studies residual instability after transcript scoring: whether plausible ASR alternatives around decision atoms would change escalation, routing, or conservative action. This makes CDS-ASR a safety layer for the remaining decision interval, not a substitute for ASR model improvement.

The aggregate-only artifact boundary is also a contribution to reproducible governance. It provides aggregate reproducibility, operation records, validation gates, dual-reviewer agreement summaries, and consistency checks under a privacy-preserving audit boundary.

The candidate lane is already bounded by evidence. Whisper large-v3, Whisper large-v3-turbo, SenseVoiceSmall, and Qwen3-ASR-0.6B have small-gate evidence but fail strict Taiwan Traditional Chinese locale promotion criteria. Qwen3-ASR-1.7B is stopped by fetch/load timeout, and Gemma 4 E2B/E4B require an isolated multimodal runtime. The next validation step is locale/runtime validation, not a broader full-split ASR run.

Supplementary Claim Registry

The claim registry is retained as supplementary governance evidence. It links paper-facing claims to aggregate artifacts, statistics, and claim-evidence alignment boundaries without taking space from the main result sequence.

Claim	Scope	Evidence artifact	Statistic	Limitation / defense
CEIS has the strongest AUC among reported predictors	Current aggregate replay surface; regenerate on completed 300-row dual-reviewer audit before final CSL submission	`human_audit_predictor_comparison.tsv`	AUC 0.9117	Decision-stability signal; full-audit ablation pending
CEIS reaches a zero-FN operating point	scoped selected-300 diagnostic threshold	`human_audit_predictor_comparison.tsv`	FN 0, recall 1.0000	Retrospective diagnostic threshold, not frozen deployment threshold
SRES and CEIS conservative policies eliminate severe misses in replay	aggregate policy replay over reviewed assessments	`policy_comparison.tsv`	high-risk missed 0, critical miss 0	Replay evidence, not live causal deployment; policies tie at 0.3889 budget
Partial encoder is the strongest current ASR hypothesis generator	258-row split/model-comparison layer	`asr_cds_proxy_comparison.tsv`	`cer_zh_micro` 15.04, unsafe down-routing 7, high-risk missed 4	Split-level model-comparison context only
Selected-300 is a high-stakes audit surface	enriched selection provenance	`human_audit_selection_summary.json`, `selection_strata.tsv`	300 candidates, 300 reviewed rows, 900 model-level assessments per reviewer	Enriched audit surface; selection provenance disclosed
Dual-reviewer audit supports label reliability	completed selected-300 review	`human_audit_reviewer_agreement_summary.json`, `human_audit_reviewer_agreement.tsv`	Cohen's kappa 0.849970–1.000000 across key fields	Transcript-bearing reviewer sheets remain local
Aggregate-only release supports reviewer-visible auditability	release and operation-record layer	validation summaries, evidence matrices, operation records, figure scripts	gate state: roadmap complete, publishable ready, consistency 26/26	Public auditability with governed local transcript-bearing evidence

Ethics, Privacy, And Intended Use

This study treats transcript-bearing speech data as sensitive operational content. NIH human-subjects guidance treats the use, study, analysis, or generation of identifiable private information as a human-subjects trigger under the cited federal definition (National Institutes of Health 2024). This paper therefore keeps raw audio, transcript-bearing hypotheses, reviewer notes, row identifiers, local response sheets, and runtime logs outside the release boundary. Before any external data release or deployment claim, the institutional review, exemption, data-use, retention, encryption, and access control status must be documented.

CDS-ASR is intended for ASR safety audit, risk-aware routing, manual-review prioritization, conservative escalation, and machine abstention. It is not intended for automatic guilt or fraud determination, automatic account freezing, service denial, punitive action, or direct law-enforcement reporting without human review. "Conservative machine action" means preserving uncertainty, raising review priority, abstaining, or requiring human confirmation.

This intended-use boundary aligns the paper with risk-management governance rather than autonomous adverse decision-making. NIST describes the AI RMF as a voluntary framework for managing AI risks to individuals, organizations, and society and for incorporating trustworthiness considerations into AI design, development, use, and evaluation (National Institute of Standards and Technology 2026). For cross-border or future deployment contexts, the EU AI Act illustrates the broader regulatory direction toward risk-based obligations and high-risk AI governance (European Commission 2026). This manuscript contributes a scoped consequence-centered audit method and a privacy-preserving release boundary that can sit inside the appropriate institutional and legal review process.

Limitations And Threats To Validity

The human-reviewed evidence now covers the full selected-300 surface with two reviewers and 900 model-level assessments per reviewer. This supports a stronger high-stakes audit and reviewer-agreement claim while leaving population-level deployment prevalence to a later study.

Model-level assessments are clustered within selected audio rows and should not be treated as independent calls. The current submission-prep package reports row-clustered bootstrap intervals and leave-one-row-out sensitivity tables for the existing predictor and recovery metrics; the full 300-row dual-reviewer audit should be used to regenerate final CSL predictor, recovery, and ablation tables.

The number of positive decision-change cases remains a key calibration handle. The final CSL table should report the positive-label count under the completed dual-reviewer audit and preserve row-clustered uncertainty for AUC and threshold behavior.

The selected-300 audit surface is deliberately enriched for high-stakes signals.

It supports decision-stability behavior within the selected audit boundary, not population prevalence of ASR-induced harm across all calls.

The reported best thresholds are diagnostic operating points on the scoped reviewed audit. They can overstate deployment performance unless threshold selection is later frozen on a separate development set.

Human labels now include a completed dual-reviewer audit and reviewer-agreement summary. Cohen's kappa is 0.849970 for decision-change, 1.000000 for semantic-risk label, 0.851426 for expected safe action, and 0.934274 for annotation confidence. The next validation layer is full-table regeneration and CEIS ablation, not additional reviewer recruitment.

Recovery evidence is aggregate-only. This protects transcript-bearing call and review content, but limits external row-level reproducibility.

CEIS depends on generated plausible variants and risk atom weights. Missed variants or misweighted atoms can affect instability scoring. The current aggregate package records risk-atom coverage and a variant-coverage status table, but a full source-specific aggregate variant generation audit remains a planned validation extension.

The Taiwan Traditional Chinese locale policy is strict by design. Candidate model rejection may reflect deployment-locale mismatch, not universal ASR inferiority.

Recovery policies eliminate high-risk missed and critical miss counts in aggregate replay, but Table 4 still leaves unsafe downrouting count at 24 after SRES/CEIS conservative recovery. The evidence supports scoped severe-miss reduction, not elimination of all safety risk.

The study evaluates whether decision-critical ASR risk can be detected and mitigated within a deliberately enriched high-stakes audit surface. A deployment trial and population-prevalence study are natural next validation layers after the completed reviewer-agreement and ablation gates are incorporated.

Appendix / Artifact Availability

Artifact Availability Statement

The repository release includes aggregate run records, validation summaries, metric tables, evidence matrices, figure-generation code, and paper-facing documentation. Raw audio, raw transcripts, selected row identifiers, audio identifiers, model hypotheses, transcript-bearing runtime logs, reviewer response sheets, reviewer notes, and model weights are not released because they may contain sensitive call or review content.

Because transcript-bearing materials may contain sensitive call content, reproducibility is provided through aggregate validation artifacts, operation records, manifest checks, and consistency audits rather than through raw row release. This provides reviewer-visible auditability for model comparison, metric policy, selected-300 human-review status, predictor analysis, recovery-policy evaluation, candidate-lane boundaries, and consistency checks while preserving the local-only boundary for sensitive materials.

The reviewer package should include an aggregate manifest with artifact path, role, privacy class, generator, source inputs, SHA256, source git commit, timestamp, Python version, metric-library versions, tokenizer policy, locale normalization policy, decoding parameters where available, random seeds where used, and hardware summary. The manifest records auditability metadata without adding transcript-bearing row content.

Reviewer-Reproducible Aggregate Artifacts

- Artifact manifest: `80_semantic_risk_asr/paper/artifact_manifest.tsv`
- Artifact manifest generator: `80_semantic_risk_asr/paper/build_artifact_manifest.py`
- Experiment registry: `70_experiments/registry.tsv`
- Main 258-row comparison: `70_experiments/runs/janus_258_test_split_asr_cds_proxy/asr_cds_proxy_comparison.tsv`
- Metric-definition audit: `70_experiments/runs/wer_metric_audit_2026_05_25/journal_compliance_summary.json`
- Selected-300 human audit refresh: `70_experiments/runs/janus_300_high_stakes_human_audit_selection_2026_05_25/human_audit_refresh_summary.json`
- Completed 300-row dual-reviewer agreement summary: `70_experiments/runs/janus_300_high_stakes_human_audit_completed_300_dual_reviewer_2026_06_02/human_audit_reviewer_agreement_summary.json`
- Completed 300-row dual-reviewer agreement table: `70_experiments/runs/janus_300_high_stakes_human_audit_completed_300_dual_reviewer_2026_06_02/human_audit_reviewer_agreement.tsv`
- Human-reviewed predictor evidence: `70_experiments/runs/janus_300_high_stakes_human_audit_selection_2026_05_25/human_audit_predictor_comparison.tsv`
- Human-reviewed predictor row-clustered CI: `70_experiments/runs/janus_300_high_stakes_human_audit_selection_2026_05_25/human_audit_predictor_clustered_ci.tsv`
- Human-reviewed predictor leave-one-row-out sensitivity: `70_experiments/runs/janus_300_high_stakes_human_audit_selection_2026_05_25/human_audit_predictor_leave_one_row_out.tsv`
- Human-reviewed recovery evidence: `70_experiments/runs/janus_300_high_stakes_recovery_human_reviewed_2026_05_26/policy_comparison.tsv`
- Human-reviewed recovery row-clustered CI: `70_experiments/runs/janus_300_high_stakes_recovery_human_reviewed_2026_05_26/policy_comparison_clustered_ci.tsv`
- Human-reviewed recovery leave-one-row-out sensitivity: `70_experiments/runs/janus_300_high_stakes_recovery_human_reviewed_2026_05_26/policy_comparison_leave_one_row_out.tsv`
- Human-reviewed recovery fixed-budget frontier: `70_experiments/runs/janus_300_high_stakes_recovery_human_reviewed_2026_05_26/fixed_budget_recovery_frontier.tsv`

- Claim registry: `70_experiments/runs/postdoc_evidence_chain_2026_05_25/claim_registry.tsv`
- CEIS method summary: `70_experiments/runs/postdoc_evidence_chain_2026_05_25/ceis_method_summary.tsv`
- Selection provenance summary: `70_experiments/runs/janus_300_high_stakes_human_audit_selection_2026_05_25/selection_provenance_summary.tsv`
- Counterfactual variant coverage status: `70_experiments/runs/janus_300_high_stakes_human_audit_selection_2026_05_25/counterfactual_variant_coverage_summary.tsv`
- Post-review evidence checklist: `70_experiments/runs/janus_300_high_stakes_human_audit_selection_2026_05_25/human_audit_post_review_evidence_summary.json`
- Publishable evidence audit: `70_experiments/runs/postdoc_evidence_chain_2026_05_25/publishable_evidence_completion_summary.json`
- Consequence evidence matrix: `70_experiments/runs/postdoc_evidence_chain_2026_05_25/consequence_evidence_matrix.tsv`
- Roadmap completion audit: `70_experiments/runs/postdoc_evidence_chain_2026_05_25/postdoc_roadmap_completion_summary.json`
- Evidence-chain consistency audit: `70_experiments/runs/postdoc_evidence_chain_2026_05_25/evidence_chain_consistency_summary.json`
- Candidate bounded recheck: `70_experiments/runs/asr_candidate_current_recheck_2026_05_26/summary.json`

Supplementary Candidate And Mechanism Figures

The candidate lane and aggregate mechanism figures are retained as supplementary evidence. They support reproducibility, scope tracing, and mechanism inspection while keeping the main article focused on method, core predictor evidence, risk mechanism, and policy replay.

Table 4: Candidate and exploratory lane evidence.

Candidate	Rows	CER	WER	Locale violations	Decision
Whisper large-v3	15	33.33	42.04	2	bounded exploratory lane
Whisper large-v3 turbo	15	40.36	50.87	4	bounded exploratory lane
SenseVoice small	15	63.12	78.98	14	bounded exploratory lane
Qwen3-ASR 0.6B	15	64.16	81.07	15	bounded exploratory lane

Candidate models are engineering-gated evidence and are not promoted to the main ASR benchmark or selected-300 claims from this table.

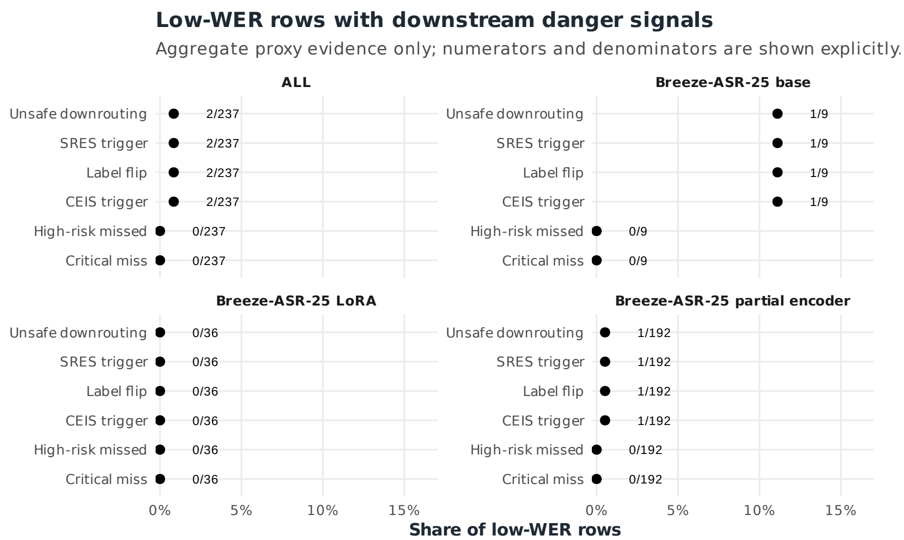

Figure 7: Low-WER danger signals under aggregate proxy evidence, with numerators and denominators shown explicitly. Source: `low_wer_danger_summary.tsv`.

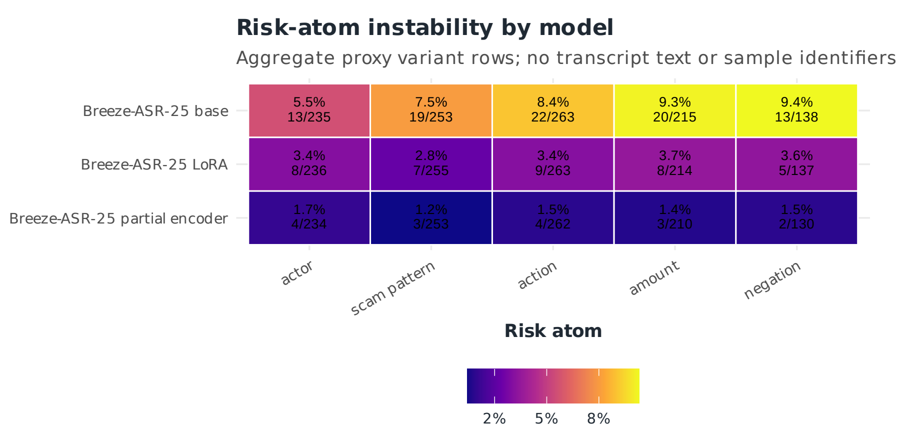

Figure 8: Risk-atom instability by model under aggregate proxy evidence. Source: `risk_atom_instability.tsv`.

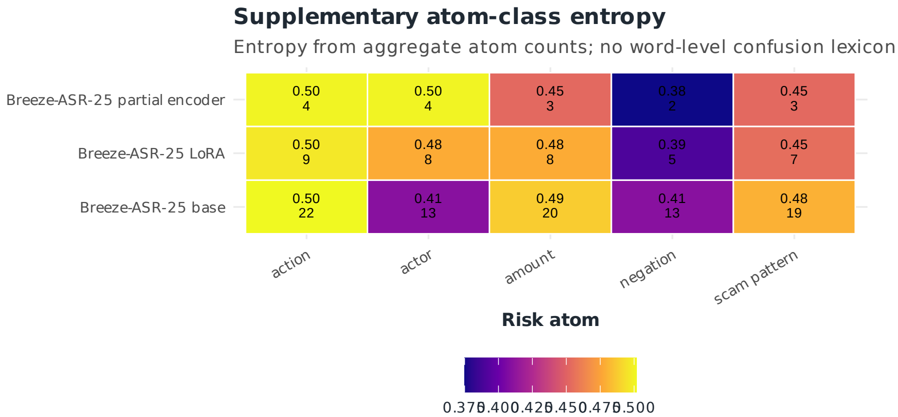

Figure 9: Atom-class entropy from aggregate instability counts. Source: `risk_atom_instability.tsv`.

Appendix Table A1. Fixed-Budget Recovery Frontier

The fixed-budget frontier is a retrospective aggregate replay over the same 90 reviewed model assessments. It reports what remains after applying ranked conservative triggers at requested trigger budgets. Requested 40% budget maps to 35 eligible triggers, so the observed budget is 0.3889.

Table 5: Fixed-budget conservative replay frontier.

Metric	Requested	Triggers	Observed	Unsafe	High-risk	Critical	Severe	Eliminated	Trig. / elim.
SRES	10%	9	10.0%	28	4	0	4	3	3.00
SRES	20%	18	20.0%	26	2	0	2	5	3.60
SRES	30%	27	30.0%	26	2	0	2	5	5.40
SRES	40%	35	38.9%	24	0	0	0	7	5.00
CEIS	10%	9	10.0%	24	0	0	0	7	1.29
CEIS	20%	18	20.0%	24	0	0	0	7	2.57
CEIS	30%	27	30.0%	24	0	0	0	7	3.86
CEIS	40%	35	38.9%	24	0	0	0	7	5.00

Retrospective aggregate replay over 90 reviewed model assessments. Requested 40% budget maps to 35 eligible triggers, so the observed budget is 38.9%.

Evidence source: `70_experiments/runs/janus_300_high_stakes_recovery_human_reviewed_2026_05_26/fixed_budget_recovery_frontier.tsv`.

Operation Records

The selected-300 reviewer route is documented through aggregate-only operation records, including response closeout, response apply logs, reviewer work order, post-review sequence, operation-record audit, and consistency audit. These records provide command-level reproducibility without exposing transcript or row content.

Key records:

- `70_experiments/runs/janus_300_high_stakes_human_audit_selection_2026_05_25/human_audit_response_closeout_summary.json`
- `70_experiments/runs/janus_300_high_stakes_human_audit_selection_2026_05_25/human_audit_batch_response_apply_log.tsv`
- `70_experiments/runs/janus_300_high_stakes_human_audit_selection_2026_05_25/human_audit_review_work_order_summary.json`
- `70_experiments/runs/janus_300_high_stakes_human_audit_selection_2026_05_25/human_audit_post_review_sequence_summary.json`
- `70_experiments/runs/janus_300_high_stakes_human_audit_selection_2026_05_25/human_audit_operation_record_summary.json`

Privacy And Local-Only Boundary

The following stay local or ignored:

- raw audio;
- raw transcripts;
- selected sample IDs and audio IDs;
- transcript-bearing audit sheets and response sheets;
- reviewer notes;
- raw model hypotheses and predictions;
- transcript-bearing runtime logs;
- checkpoints, adapters, and model weights;
- local downloaded reviewer packets and zip files.

The tracked repo preserves aggregate counts, run records, validation summaries, metric tables, and paper-facing evidence matrices. This boundary lets reviewers audit the evidence chain while protecting transcript-bearing material.

Validation Gate Commands

Run these after manuscript packaging changes:

```
.venv/bin/python 80_semantic_risk_asr/scoring/audit_postdoc_roadmap_completion.py --output-j
.venv/bin/python 80_semantic_risk_asr/scoring/audit_publishable_evidence_chain.py --output-j
.venv/bin/python 80_semantic_risk_asr/scoring/audit_evidence_chain_consistency.py --output-j
```

Expected current gate state:

- roadmap completion: `roadmap_complete=true`, `blocking_gate=none`;
- publishable evidence: `publishable_ready=true`;
- consistency audit: `26/26` checks pass, `failed_checks=[]`.

Scope Control For Additional Experiments

Do not run new full-split ASR experiments now. Candidate models can move to 258-row or selected-300 only after strict Taiwan Traditional Chinese locale policy is satisfied or an isolated Gemma 4 multimodal runtime exists. Until then, the active work is manuscript drafting, results-table packaging, artifact availability, citation completion, and submission readiness.

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